

3.8 The Root Test

At times the ratio test will be difficult to use since it requires the calculation of a limit that may be difficult to determine. In such situations this next test can be useful.

Theorem (The Root Test): Consider the series $\sum_{k=1}^{\infty} a_k$ and let $\rho = \lim_{k \rightarrow \infty} \sqrt[k]{|a_k|} = \lim_{k \rightarrow \infty} (|a_k|)^{1/k}$.

1. If $\rho < 1$, the series converges absolutely.
2. If $\rho > 1$ or $\rho = +\infty$, the series diverges.
3. If $\rho = 1$, the test is inconclusive and no conclusion can be drawn.

We will omit the proof of this theorem.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \frac{1}{(\ln(k+1))^k}$ converges or diverges.

Example 2: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{-2n}{n+1} \right)^{5n}$ converges or diverges.

Example 3: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \left(\frac{4k-5}{2k+1} \right)^k$ converges or diverges.